

6. Safety Concept, Traceability and Verification

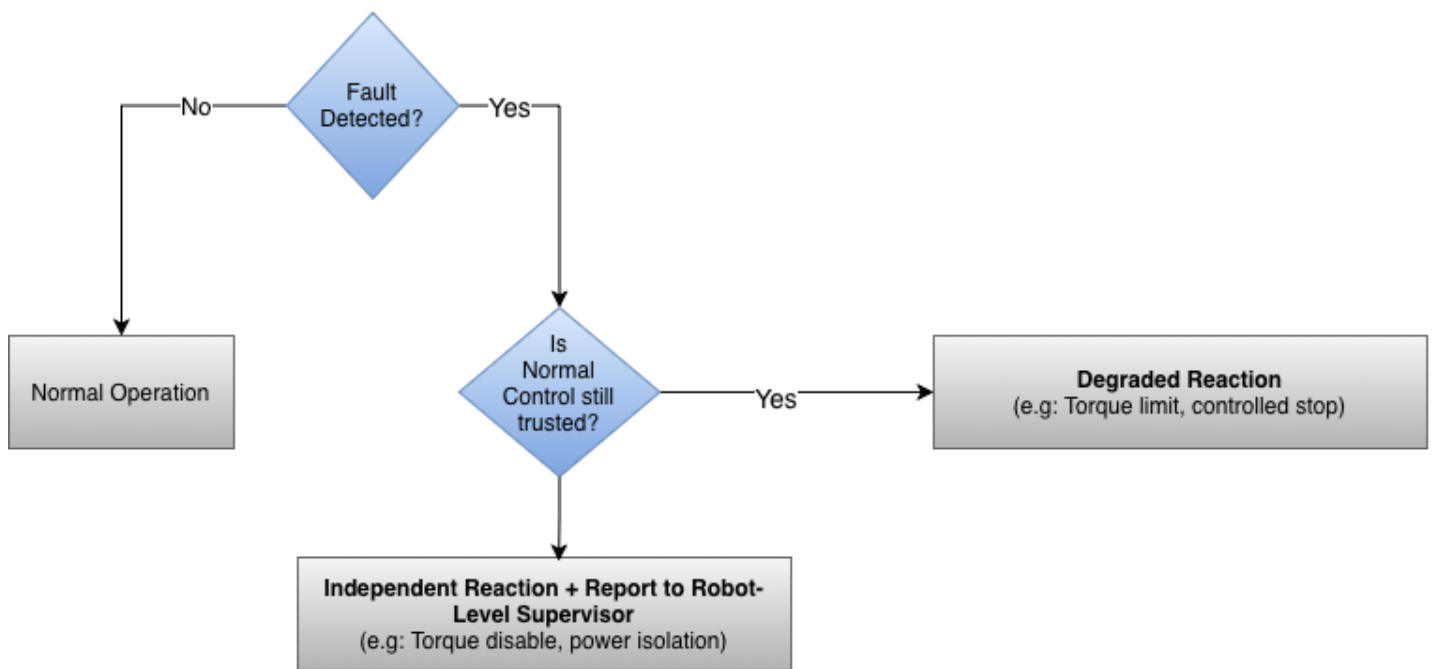
6.1 Purpose

This section defines the fault-reaction concept and links the safety requirements to architecture and verification.

Safety focus: HE-01 - hazardous unintended knee motion while the joint is load-bearing. All concepts below trace back to SG-01 and the ten FSRs defined earlier.

6.2 Safety Concept and Fault-Reaction Logic

Figure 6 - Fault-Reaction Decision Concept



The safety concept is based on the amount of control capability that remains trustworthy after a fault. A communication fault, sensor fault and inverter fault therefore do not automatically result in the same reaction.

If local control and sensing remain trustworthy, the preferred response is to restrict capability and move toward a controlled stop. If the normal control or torque-generation path can no longer be trusted, the reaction must use an independent mechanism.

Immediate torque-off is not treated as the default response because loss of knee support may itself destabilise the robot.

6.3 Fault-Reaction Matrix

Detected condition	Trusted capability	Reaction
Communication unavailable / invalid	Local control and sensing healthy	Controlled stop, then local hold / degraded state
Safety-relevant sensor fault	Remaining sensing and controller healthy	Restrict torque/velocity; degraded mode or controlled stop
Joint motion / transmission inconsistency	Control responsive, mechanical state uncertain	Torque limitation + controlled stop / robot-level recovery
Local joint controller untrusted	Independent safety path healthy	Independent torque disable
Gate driver / inverter untrusted	Safety monitor and independent isolation healthy	Gate disable / power isolation + coordinated recovery
Safety path itself unavailable	Normal control healthy but protection degraded	Prevent unrestricted operation and perform controlled robot-level stop

6.4 Independent Safety Path

The normal controller may itself be the source of hazardous torque. The architecture therefore includes a hardware-capable reaction path that can inhibit torque generation without relying solely on the normal control software.

The final implementation must address common dependencies such as shared power, processor, clock and gate-control resources.

6.5 Requirements-to-Architecture Traceability

The table below shows how each functional safety requirement is implemented conceptually in the architecture and how it will be verified. HE-01 / SG-01 apply to all rows.

FSR	Architecture Element	Safety Mechanism	V&V Ref
FSR-01	Joint Encoder + Safety Monitor	Velocity-deviation monitoring	VVT-01
FSR-02	Safety Monitor + Motor Control	Torque/velocity limitation	VVT-01
FSR-03	Safety Monitor + Communication Interface	Fault-status reporting	VVT-07
FSR-04	Current Sensors + Safety Monitor	Current-deviation monitoring	VVT-02
FSR-05	Safety Monitor + Gate/Disable Path	Current limitation / torque inhibit	VVT-02
FSR-06	Motor Encoder + Joint Encoder	Transmission plausibility	VVT-03

FSR	Architecture Element	Safety Mechanism	V&V Ref
FSR-07	Communication Interface	Timeout / integrity supervision	VVT-04
FSR-08	Local Controller + Safety Monitor	Invalid-command rejection / controlled reaction	VVT-04
FSR-09	Independent Torque-Disable Path	Independent torque inhibit	VVT-06
FSR-10	Sensors + Safety Monitor	Sensor plausibility / degraded reaction	VVT-05

6.6 Verification Strategy

Verification is planned progressively. SiL/HiL testing is used for diagnostic logic and fault injection; actuator bench testing covers the real motor, inverter and sensors; integration testing covers interaction with the robot-level safety supervisor. Analysis is used for timing, independence and common-cause assumptions

6.7 V&V Matrix

V&V ID	Verification objective	Requirements	Method	Pass concept
VVT-01	Joint velocity deviation monitoring and reaction	FSR-01, FSR-02	SIL / HiL fault injection	Inject excessive commanded-vs-measured velocity deviation; detect and initiate defined reaction within required time.
VVT-02	Motor current monitoring and reaction	FSR-04, FSR-05	HiL / bench test	Inject incorrect current feedback or emulate overcurrent; detect and limit / disable as defined.
VVT-03	Motor-to-joint transmission plausibility	FSR-06	SIL / HiL	Inject encoder offset or simulated gearbox slip; detect mismatch above allowable tolerance.
VVT-04	Communication supervision and reaction	FSR-07, FSR-08	Communication fault injection	Drop, delay, repeat and corrupt commands; reject invalid data and enter defined reaction.
VVT-05	Sensor fault handling	FSR-10	SIL / HiL fault injection	Freeze, offset or remove safety-relevant sensor signals; prevent unrestricted actuator operation.
VVT-06	Independent safety reaction path	FSR-09	HiL / integration	Force a main-controller fault while hazardous torque is requested; verify independent path still inhibits torque.

V&V ID	Verification objective	Requirements	Method	Pass concept
VVT-07	Fault reporting to robot safety supervisor	FSR-03	Integration test	Inject representative safety faults; verify correct fault/status message arrives within required reporting time.

6.8 Open Points and Limitations

- Diagnostic thresholds, persistence times and reaction-time budgets still need quantitative derivation.
- Safety Monitor independence and the torque-disable implementation require detailed hardware/software design.
- Quantitative reliability, diagnostic coverage and common-cause analysis are outside the current demonstrator scope.
- Verification activities are planned; physical test evidence has not yet been generated.